Cross-Attention Spectral–Spatial Network for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image (HSI) classification aims to identify categories of hyperspectral pixels. Recently, many convolutional neural networks (CNNs) have been designed to explore the spectrums and spatial information of HSI for classification. In recent CNN-based methods, 2-D or 3-D convolutions are inevitably utilized as basic operations to extract the spatial or spectral–spatial features. However, 2-D and 3-D convolutions are sensitive to the image rotation, which may result in that recent CNN-based methods are not robust to the HSI rotation. In this article, a cross-attention spectral–spatial network (CASSN) is proposed to alleviate the problem of HSI rotation. First, a cross-spectral attention component is proposed to exploit the local and global spectrums of the pixel to generate band weight for suppressing redundant bands. Second, a spectral feature extraction component is utilized to capture spectral features. Then, a cross-spatial attention component is proposed to generate spectral–spatial features from the HSI patch under the guidance of the pixel to be classified. Finally, the spectral–spatial feature is fed to a softmax classifier to obtain the category. The effectiveness of CASSN is demonstrated on three public databases.**

***Index Terms*— Convolutional neural networks (CNNs), hyperspectral image (HSI) classification, spatial attention, spectral attention.**

I. Introduction

HYPERSPECTRAL imaging instruments can collect hundreds of continuous electromagnetic spectral bands from observed remote sensing scenes [1], [2]. Since the rich spectrums can reveal intrinsic physical and chemical properties of objects [3], [4], hyperspectral image (HSI) is conducive to distinguishing different materials contained in remote sensing scenes. HSI classification refers to identifying the land-cover categories of pixels [5], which contributes to various applications, such as disaster monitoring, urban planning, and intelligent agriculture [6]–[8].

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HSI is the intrinsic 3-D data that consists of 1-D spectrums and 2-D spatial information, both of which contain valuable clues for HSI classification [9]. To explore the 1-D spectrums and 2-D spatial information for HSI classification, recent works have built plentiful convolutional neural networks (CNNs) that perform better than previous handcrafted-feature-based methods [10], [11]. According to the structural characteristics of deep networks, recent CNN-based methods can be summarized into pixelwise methods, patchwise methods, and pixel-patchwise methods.

Pixelwise methods treat the HSI as a set of pixels and directly take the pixel as the input to fully explore its 1-D spectrums for identifying the category of the corresponding land cover [12], [13]. In pixelwise methods, 1-D CNNs [14], [15] have shown powerful spectral feature extraction (SFE) capabilities for HSI classification. Since pixelwise methods only explore the spectrums of pixels in the spectral domain, these methods are robust to the HSI rotation in the spatial domain [16]. However, pixelwise methods ignore the 2-D spatial information contained in the HSI [17].

Patchwise methods aim to exploit the 2-D spatial information of HSI to boost the classification performance [18], [19]. To leverage the 2-D spatial information, patchwise methods usually assume that neighboring pixels have the same category based on the spatial smoothness prior [11]. The HSI patch is cropped from the original HSI as the input of patchwise methods. Then, many well-designed 2-D CNNs and 3-D CNNs have shown powerful spatial feature extraction capabilities for HSI classification [20], [21]. In general, patchwise methods transform the hyperspectral pixel classification problem into the HSI patch classification problem [22] and equate the category of the HSI patch to the category of the center pixel [9]. However, patchwise methods mainly focus on exploiting the spatial features of the whole HSI patch [23], [24] and cannot fully explore the 1-D spectrums of the center pixel (referring to the pixel to be classified) in the HSI patch. In addition, there will be different categories of pixels in the HSI patch, which can weaken the discrimination of spatial features [25].

Pixel-patchwise methods attempt to simultaneously explore the 1-D spectrums of the pixel to be classified and the 2-D spatial information of the HSI patch for HSI classification [12], [26]. Different from pixelwise methods and patchwise methods that are single-branch networks, pixel-patchwise methods are two-branch networks. One branch takes the pixel to be classified as the input to extract spectral features, and the other branch takes the HSI patch as the input to extract spatial

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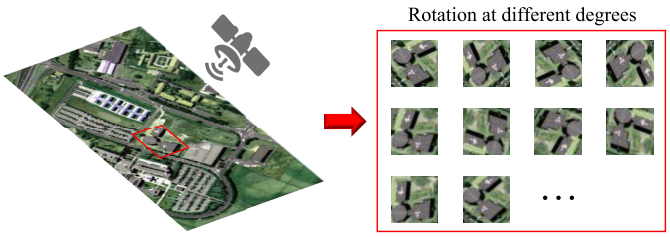


Fig. 1. Changes of imaging perspectives can easily cause the spatial rotation of the captured HSIs.

features [27]. Then, spectral features and spatial features are aggregated for HSI classification.

In both patchwise methods and pixel-patchwise methods, the 2-D/3-D convolutions are widely utilized as basic operations to extract the spatial or spectral–spatial features from the HSI patch. In these methods, the spatial size of kernels in 2-D/3-D convolutions is usually set to 3×3 to leverage the spatial information of the HSI patch [12]. However, the 2-D/3-D convolutions with 3×3 kernels are not robust to the image rotation [28], which will cause that recent patchwise methods and pixel-patchwise methods are not robust to the HSI rotation [29]. Due to the unique bird's-eye perspective in the remote sensing hyperspectral imaging process, the captured HSIs are prone to rotation [30]. As shown in Fig. 1, the land covers in rotated images may appear in any orientation. The rotation variability of the captured HSIs can make the HSI patch easy to rotate, which further causes the performance of patchwise methods and pixel-patchwise methods to decrease.

To alleviate the problem that previous patchwise methods and pixel-patchwise methods are not robust to HSI rotation, this article proposes a cross-attention spectral–spatial network (CASSN). The proposed CASSN belongs to the pixel-patchwise methods. The structures of CASSN are composed of the cross-spectral attention (CEA) component, the SFE component, the cross-spatial attention (CAA) component, and the classifier component. First, the CEA component is proposed to simultaneously suppress the redundant bands contained in the input pixel and the input HSI patch. The local information and global information of spectrums in the input pixel are explored to generate the band weight, and then, the band weight is utilized to simultaneously recalibrate the importance of spectral bands in the input pixel and the input HSI patch. Second, the SFE component is designed to capture features from the recalibrated pixel and HSI patch in the spectral dimension. Since both the pixel and HSI patch contain plentiful spectral bands, it is necessary to extract spectral features to make full use of these spectral bands. Then, the CAA component is proposed to replace 2-D and 3-D convolutions to generate rotation-invariant spectral–spatial features. The proposed CAA component first captures the correlation between the neighboring pixels in the input HSI patch and the input pixel. Then, the correlation is utilized to aggregate the neighboring pixels in the input HSI patch to generate rotation-invariant spectral–spatial features. Finally, the spectral–spatial features are fed to the softmax classifier to obtain the category. Three public databases are utilized to demonstrate the effectiveness of CASSN.

The contributions of this article are given as follows.

1) A CASSN is proposed to alleviate the HSI rotation problem. The proposed CASSN contains a pixel branch and a patch branch to fully explore the 1-D spectrums and 2-D spatial information of HSI for classification.
2) A cross-spectral attention module (SAM) is proposed to suppress redundant bands. The proposed cross-SAM generates the band weight based on the input pixel, which can avoid interference from other categories of pixels. Then, the band weight is utilized to simultaneously recalibrate the bands of the input pixel and the HSI patch.
3) A CAA module is proposed to generate rotation-invariant spectral–spatial features. The proposed CAA module can adaptively aggregate spectral features from neighboring pixels into the spectral–spatial features.

The rest sections are organized as follows. First, recent works are reviewed in Section II. Then, Section III introduces the details of CASSN. Next, Section IV gives the experimental settings and results. Finally, Section V summarizes this article.

II. Related Work

In hyperspectral remote sensing, the rich spectrums of HSI provide great potential to accurately identify the categories of land covers [31], [32]. To identify the categories of hyperspectral pixels, previous methods usually focus on designing various feature extraction methods based on low-rank and sparse priors of HSI [33], [34]. For example, Rasti *et al.* [35] explored the low-rank and spatial smoothness priors of HSI to extract features. Luo *et al.* [36] utilized the sparse-adaptive hypergraph to reveal the structure relationships among pixels. Peng *et al.* [37] combined self-paced learning strategy with sparse representation framework to improve the robustness to noise. A detailed review to feature extraction methods can be found in literature [11], [33].

Recently, many deep learning-based methods have been proposed for HSI classification. Various deep networks, such as stacked autoencoder networks [13], [38], recurrent neural networks [39]–[41], and graph neural networks (GNNs) [42], [43], have been extensively studied in HSI classification. Based on the differences in the network structures, recent deep learning-based methods can be roughly divided into pixelwise methods, patchwise methods, and pixel-patchwise methods.

A. Pixelwise Methods

Pixelwise methods aim to fully exploit the 1-D spectrums of the pixel to obtain its category. 1-D CNN can effectively explore the relationship among local adjacent bands and is widely used in pixelwise methods to extract spectral features for HSI classification [40], [44]. Hu *et al.* [44] proposed a plainly five-layer 1-D CNN to explore the 1-D spectrums of the hyperspectral pixel for HSI classification. Li *et al.* [15] proposed a pixel-pair method based on the 1-D CNN to mitigate the problem of insufficient samples. Wu and Prasad [40] integrated 1-D CNN and recurrent neural networks into a unified structure for HSI classification.

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HSI contains hundreds of bands that have high information redundancy. Suppressing redundant bands can effectively improve the discrimination of the spectrums of the pixel [6]. Inspired by the human visual attention, Li *et al.* [14] proposed a spectral feature fusion network to extract the interlocal and local features from the 1-D spectrums. In addition, the nonlocal and global channel attention modules are designed to exploit the nonlocal and global channel correlations to improve the discrimination of features.

Since HSI is 3-D data, its 2-D spatial information can also provide effective clues for HSI classification [16], [45]. Pixelwise methods regard the HSI as a collection of pixels. Pixel-based methods only leverage the 1-D spectrums of the pixel and fail to explore the 2-D spatial information of HSI, resulting in poor performance [11]. However, precisely because pixelwise methods do not use the 2-D spatial information of HSI, HSI rotation has little effect on the performance of these methods. Pixelwise methods can fully explore the 1-D spectrums of the pixel to be classified and are robust to HSI rotation.

B. Patchwise Methods

Patchwise methods mainly explore the 2-D spatial information of the cropped HSI patch to boost the classification results [18], [19]. 2-D or 3-D convolutions are utilized as basic operations to extract spatial or spectral–spatial features from HSI patches [12]. Since the HSI patch is composed of various neighboring pixels, the HSI patch actually contains both the spectrums of neighboring pixels and the spatial information. Many patchwise methods mainly focus on exploiting the spatial information and whole spectrums of neighboring pixels for HSI classification. Chen *et al.* [46] proposed a plainly 2-D CNN to exploit the HSI patch for classification. Cao *et al.* [17] combined the 2-D CNN with Markov random fields to generate smooth classification maps. He *et al.* [47] proposed to utilize the 2-D CNN to extract discriminative features from multiscale covariance maps. Zhong *et al.* [48] proposed a 3-D residual network to capture spectral–spatial features. However, when the HSI patch is close to the region edges, the HSI patch will contain other categories of pixels, which can weaken the discrimination of the global spatial information in the HSI patch [25]. In addition, the redundant bands in the HSI patch will also reduce the classification results [45], [49].

Recent works have proposed various spatial attention components to alleviate the interference from other categories of pixels. In addition, various spectral attention components are designed to alleviate the problem of band redundancy. Fang *et al.* [23] proposed a 3-D dense convolutional network to extract multiscale spatial features. Spectral attention is proposed to improve the distinguishability of the multiscale spatial features. Wang *et al.* [45] employed a neural architecture search method to find a lightweight spectral–spatial network for HSI classification. The efficient channel attention is utilized to adjust the weights of spectral bands. Mou and Zhu [49] proposed spectral attention to suppress the importance of redundant bands. Zhu *et al.* [50] proposed a spectral–spatial attention to capture the informative spectral bands and spatial regions from the raw HSI patch. Then, residual convolution blocks are utilized to extract discriminative features. Xue *et al.* [51] proposed a second-order pooling operation based on the attention mechanism to refine the spectral–spatial features. Lin *et al.* [52] embedded the attention mechanism into the pseudo-3-D network for HSI classification. Zhang *et al.* [53] first divided the spectral bands into nonoverlapped subbands and then utilized parallel CNNs to extract features. In addition, a homogeneous pixel detection module is proposed to mitigate the influence of interfering pixels. Spectral attention and spatial attention have been extensively studied in the HSI classification and can effectively improve classification results. However, the patchwise methods tend to focus too much on the use of the spatial information of neighboring pixels, which easily leads to the underutilization of the spectrums of the center pixel (referring to the pixel to be classified).

C. Pixel-Patchwise Methods

Pixel-patchwise methods can combine the advantages of both pixelwise methods and patchwise methods. The 1-D spectrums of the pixel and the 2-D spatial information of the HSI patch can be exploited for HSI classification. Yang *et al.* [54] proposed the recurrent 2-D/3-D CNNs to extract spectral–spatial features from multilevel HSI patches. Zhou *et al.* [55] proposed to utilize long short-term memory (LSTM) to extract spectral features from the spectrums of the pixel and extract spatial features from neighborhood regions. Hao *et al.* [27] proposed a two-branch network, where one branch is the stacked autoencoder network for extracting spectral features from the pixel and the other branch is the 2-D CNN for extracting spatial features from the HSI patch. Li *et al.* [26] proposed a two-branch network, which is composed of two 2-D CNN. The channel correlation is exploited to extract the informative features. Hong *et al.* [56] proposed a two-branch fusion network, where one branch is a graph convolutional network that can model long-range spatial relations, and the other is the traditional CNN that is utilized to extract features from short-range regions.

These methods mainly have two problems. First, these methods ignore the redundant bands in the spectrums of the pixel and directly extract spectral features from the spectrums of the pixel. However, redundant bands will reduce the discrimination of spectral features. Second, the 2-D or 3-D convolutions with 3×3 kernels are still utilized to extract spectral–spatial features from the HSI patch. However, the 2-D or 3-D convolutions with 3×3 kernels are sensitive to the image rotation [30], which will cause that these methods are not robust to the HSI rotation [29].

III. Proposed Method

This article proposes a CASSN to alleviate the problem of HSI rotation. First, Section III-A introduces the overall structures of CASSN. Second, the proposed CEA component is described in Section III-B. Then, the SFE component is

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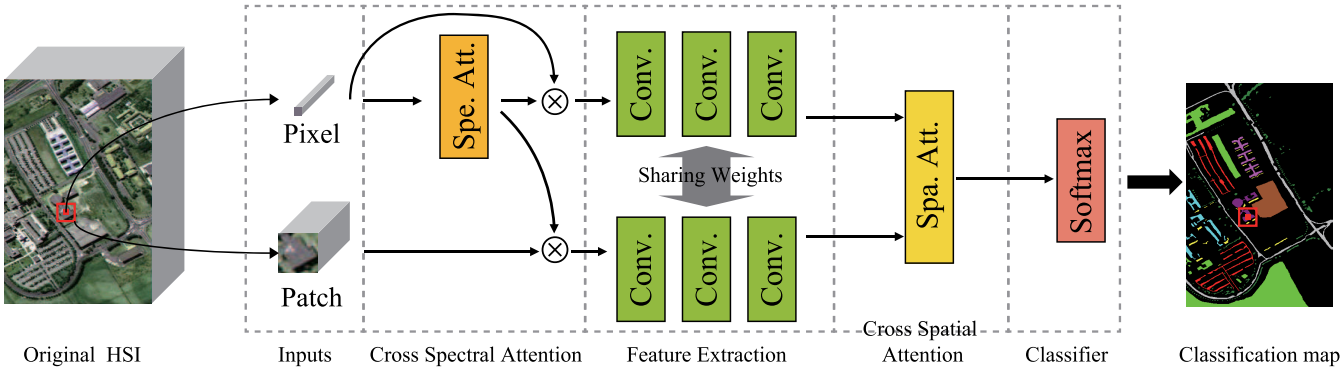


Fig. 2. Overall structures of CASSN are composed of the CEA component, the SFE component, the CAA component, and the classification component. ⊗ represents the bandwise multiplication.

introduced in Section III-C. Finally, Section III-D gives the details of the proposed CAA component.

A. Overview

Let $\mathcal{H} \in \mathbb{R}^{M\times N\times B}$ represent the original 3-D HSI, where $M \times N$ is the spatial size of the HSI $\mathcal{H}$ and B is the band number of the HSI $\mathcal{H}$. HSI classification aims to obtain categories of pixels in the HSI $\mathcal{H}$. In the proposed CASSN, the pixel $p_i \in \mathbb{R}^{1\times 1\times B}$ to be classified is first split from HSI $\mathcal{H}$ to provide the spectrums, and the HSI patch $H_i \in \mathbb{R}^{q\times q\times B}$ is cropped from HSI $\mathcal{H}$ to provide the spatial information, where $q \times q$ is the spatial size of the HSI patch H_i. The pixel p_i uniquely corresponds to the HSI patch H_i, and the center pixel of the HSI patch H_i is the pixel p_i. Both the pixel p_i and the HSI patch H_i are exploited as the inputs of CASSN for HSI classification.

The overall structures of CASSN are shown in Fig. 2. The proposed CASSN is a two-branch network, where one branch is the pixel branch that takes the pixel p_i as the input, and the other is the patch branch that takes the HSI patch H_i as the input. Different from the previous two-branch pixel-patchwise methods that two branches extract features independently [27], the pixel branch and the patch branch in the proposed CASSN are interactive to generate spectral–spatial features.

The proposed CASSN is composed of four primary components: the CEA component, the SFE component, the CAA component, and the classification component. The proposed CEA component aims to suppress the redundant spectral bands. Different from previous methods that suppress the redundant spectral bands of the HSI patch H_i based on the spectrums of whole neighboring pixels in the HSI patch H_i, the proposed CEA component exploits the spectrums of the pixel p_i to simultaneously suppress the redundant spectral bands contained in the pixel p_i and the HSI patch H_i. The SFE component aims to extract features in the spectral dimension. The SFE component contains two subnetworks, where one subnetwork for extracting spectral features from the pixel p_i and the other for extracting spectral features from the HSI patch H_i. The proposed CAA component aims to aggregate the spectral features of neighboring pixels in the HSI patch H_i under the guidance of the spectral features of the pixel p_i to generate the spectral–spatial features. The classification component aims to infer the category of the pixel p_i based on the obtained spectral–spatial features. The cross-entropy loss is employed as the loss function of CASSN for training. The network structures of CASSN are listed in Table I. The details of the CEA component, the SFE component, and the CAA component are introduced as follows.

B. CEA Component

The HSI contains rich spectral bands that are critical for identifying the categories of pixels [57], [58]. Recent works have demonstrated that different spectral bands contribute differently to the HSI classification task [59], [60]. Therefore, highlighting informative spectral bands or suppressing redundant spectral bands can significantly improve the results of HSI classification [49]. To highlight informative spectral bands and suppress redundant spectral bands in a task-driven manner, many works have built various spectral attention components that can be embedded into deep networks for end-to-end training [19], [23], [50]. The main purpose of the spectral attention component is to generate the band weight $W_i \in \mathbb{R}^B$ to recalibrate the importance of each spectral band. Previous methods usually generate the band weight W_i based on the whole neighboring pixels in the input HSI patch H_i to suppress redundant spectral bands of the HSI patch H_i [45], [50], [61]. However, when the input HSI patch H_i is located near the land-cover region edges [19], [25], the HSI patch H_i usually contains other categories of pixels. Since different categories of pixels have different redundant spectral bands [49], other categories of pixels can interfere with the band weight W_i.

In this article, a CEA component is proposed to fully explore the input pixel p_i to generate the band weight W_i. Then, spectral bands of the pixel p_i and the HSI patch H_i are simultaneously recalibrated by the band weight W_i. The proposed CEA component only uses the pixel p_i to generate the band weight W_i, which can effectively avoid the interference of other categories of pixels in the HSI patch H_i.

In the CEA, both local information and global information of spectrums in the pixel p_i are explored to generate the band weight W_i. Since the HSI is obtained by intensive sampling along with the electromagnetic signal, local adjacent bands in spectrums are highly correlated [57]. To explore the local adjacent band correlation of spectrums, the 1-D convolution operation with the sigmoid activation is employed

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TABLE I
DETAILED STRUCTURES OF THE PROPOSED CASSN

	Pixel Branch		Patch Branch
Input	Pixel p_i		HSI Patch H_i
Cross Spectral Attention Component	Layer: 1-D Convolution with Sigmoid Input: p_i. Output: W_i^{local}.	Layer: Fully Connection with Sigmoid Input: p_i. Output: W_i^{global}.	
	Layer: Hadamard Product Input: W_i^{local} and W_i^{global}. Output: W_i.		
	Layer: Band-Wise Product Input: W_i and p_i. Output: $\bar{p}_i$.		Layer: Band-Wise Product Input: W_i and H_i. Output: $\bar{H}_i$.
Spectral Feature Extraction Component	Layer: 1×1 2-D Convolution with ReLU Input: $\bar{p}_i$. Output: Conv-Spe-1.		Layer: 1×1 2-D Convolution with ReLU Input: $\bar{H}_i$. Output: Conv-Spa-1.
	Layer: 1×1 2-D Convolution with ReLU Input: Conv-Spe-1. Output: Conv-Spe-2.		Layer: 1×1 2-D Convolution with ReLU Input: Conv-Spa-1. Output: Conv-Spa-2.
	Layer: 1×1 2-D Convolution with ReLU Input: Conv-Spe-2. Output: F_{pixel}.		Layer: 1×1 2-D Convolution with ReLU Input: Conv-Spa-2. Output: F_{patch}.
Cross Spatial Attention Component	Layer: 1×1 2-D Convolution with ReLU Input: F_{pixel}. Output: $\bar{F}_{pixel}$.		Layer: 1×1 2-D Convolution with ReLU Input: F_{patch}. Output: $\bar{F}_{patch}$.
	Layer: Product Input: $\bar{F}_{pixel}$ and $\bar{F}_{patch}$. Output: R.		
	Layer: Normalization Input: R. Output: $\bar{R}$.		
	Layer: Product Input: $\bar{R}$ and $\bar{F}_{patch}$. Output: $F_{spe-spa}$.		
Classification	Layer: Softmax Classifier Input: $F_{spe-spa}$. Output: Predicted category.		

to generate the local band weight W_i^{local}. Let $W_i^{\text{local}}(j)$ denote the jth element of W_i^{local}. The value of $W_i^{\text{local}}(j)$ indicates the importance of the jth band in the pixel p_i based on the local spectrums. $W_i^{\text{local}}(j)$ is obtained based on m local adjacent bands, and the formulation of $W_i^{\text{local}}(j)$ is given as follows:

$$W_i^{\text{local}}(j) = \varphi\left(\sum_{k=1}^{m} \theta_l(k) p_{i,j}(k)\right) \tag{1}$$

where $p_{i,j}(k)$ represents the kth local adjacent bands of the jth band in the pixel p_i, m is the number of total local adjacent bands, φ is the sigmoid activation function, $\theta_l \in \mathbb{R}^m$ is the weight of the 1-D convolution, and the bias is omitted. The value of m is empirically set to 7. In addition, the fully connected layer with the sigmoid activation is utilized to explore the global spectrums of the pixel p_i for generating the global band weight $W_i^{\text{global}} \in \mathbb{R}^B$. The formulation of W_i^{global} is given as follows:

$$W_i^{\text{global}} = \varphi(\theta_g p_i) \tag{2}$$

where $\theta_g \in \mathbb{R}^{B \times B}$ is the weight of this fully connected layer, and the bias is also omitted. The value of $W_i^{\text{global}}(j)$ indicates the importance of the jth band in the pixel p_i based on the global spectrums. When both local information and global information of spectrums consider the jth band in the pixel p_i to be important for classification, this jth band in the pixel p_i is the informative band; otherwise, it is the redundant band. The final importance $W_i(j)$ of the jth band in the pixel p_i is jointly determined by $W_i^{\text{local}}(j)$ and $W_i^{\text{global}}(j)$

$$W_i(j) = W_i^{\text{local}}(j) W_i^{\text{global}}(j). \tag{3}$$

Finally, the band weight W_i is utilized to simultaneously recalibrate the importance of different bands in the pixel p_i and the HSI patch H_i

$$\bar{p}_i = W_i \otimes p_i \tag{4}$$

$$\bar{H}_i = W_i \otimes H_i \tag{5}$$

where $\bar{p}_i \in \mathbb{R}^B$ is the recalibrated pixel, $\bar{H}_i \in \mathbb{R}^{q \times q \times B}$ is the recalibrated HSI patch, and $\otimes$ is the bandwise multiplication. Then, the recalibrated pixel $\bar{p}_i$ and the recalibrated HSI patch $\bar{H}_i$ are fed to the SFE component for obtaining spectral features.

C. SFE Component

The SFE component is designed to extract features in the spectral dimension. Since the HSI patch H_i is composed of various neighboring pixels, the HSI patch should also be fed to the SFE component. As shown in Fig. 2, the SFE component consists of two subnetwork. One subnetwork extracts spectral features $F_{\text{pixel}} \in \mathbb{R}^{1 \times 1 \times C}$ from the recalibrated pixel $\bar{p}_i$ in the pixel branch, and the other subnetwork extracts spectral features $F_{\text{patch}} \in \mathbb{R}^{q \times q \times C}$ from the recalibrated HSI patch $\bar{H}_i$ in the patch branch, where C is the channel number. Since both two subnetworks focus on extracting features from spectrums, the structures and the weight parameters of these subnetworks are set to be shared in training and testing

$$F_{\text{pixel}} = \text{SFE}(\bar{p}_i) \tag{6}$$

$$F_{\text{patch}} = \text{SFE}(\bar{H}_i) \tag{7}$$

where SFE$(\cdot)$ represents the designed spectral feature extraction component. In this article, the subnetworks in SFE component are composed of three 2-D convolutional layers connected in series, where the spatial size of kernels is 1×1 and the number of kernels is 64.

D. CAA Component

Recent works also design many spatial attention modules, which are designed to capture the attention regions from the HSI patch H_i [25]. In previous works, the spatial attention modules often need to be further combined with the 3×3 2-D/3-D convolutions to extract spectral–spatial features from

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the attention regions [50], [52]. However, the 3×3 2-D/3-D convolutions are sensitive to the image rotation [28], which will cause that recent methods are also sensitive to the HSI rotation [29].

Different from the previous patch- and pixel-patch-based methods that still utilize the 2-D or 3-D convolutions to extract the spatial or spectral–spatial features, this article proposes a CAA component to replace the 2-D and 3-D convolutions for the spectral–spatial feature extraction based on the spectral feature F_{pixel} of the pixel p_i and the spectral feature F_{patch} of the HSI patch H_i.

The proposed CAA component aims to fully exploit the correlation between the neighboring pixels in the HSI patch H_i and the pixel p_i to generate the spectral–spatial features. First, the 2-D convolutions with 1×1 kernels are utilized to transform F_{pixel} and F_{patch} into $\bar{F}_{\text{pixel}} \in \mathbb{R}^{1\times 1\times C_a}$ and $\bar{F}_{\text{patch}} \in \mathbb{R}^{q\times q\times C_a}$, respectively, where C_a is the channel number. Second, to facilitate the calculation of the correlation between the neighboring pixels in the HSI patch H_i and the pixel p_i, the size of $\bar{F}_{\text{pixel}}$ is reshaped to $\mathbb{R}^{1\times C_a}$, and the size of $\bar{F}_{\text{pixel}}$ is reshaped to $\mathbb{R}^{q^2\times C_a}$. The correlation $R \in \mathbb{R}^{1\times q^2}$ between the pixel p_i and the neighboring pixels in the HSI patch H_i can be calculated as

$$R = \bar{F}_{\text{pixel}} \bar{F}_{\text{patch}}^{T}. \tag{8}$$

Next, the correlation R is normalized by the softmax for generating the spectral–spatial feature

$$\bar{R} = \frac{e^{R(i)}}{\sum_{j=1}^{q^2} e^{R(j)}}. \tag{9}$$

Finally, the spectral–spatial feature $F_{\text{spe-spa}} \in \mathbb{R}^{1\times C_a}$ is obtained by aggregating the neighboring pixels in the HSI patch H_i with the normalized correlation $\bar{R}$

$$F_{\text{spe-spa}} = \bar{R}\bar{F}_{\text{patch}}. \tag{10}$$

After the CAA component, the spectral–spatial feature $F_{\text{spe-spa}}$ is fed to the softmax classifier to obtain the category.

IV. Experiments

A. Databases and Evaluation Measures

To demonstrate the performance of CASSN, three public databases, i.e., Pavia University (PU), Indian Pines (IP), and Salinas,[1] are utilized to conduct experiments.

The PU database was obtained by the reflective optics system imaging spectrometer (ROSIS) instrument over Pavia. This database consists of 610×340 pixels, and each pixel has 103 bands. The spectral wavelength ranges from 430 to 860 nm. This database is interested in nine categories of common land covers on the campus. Table II lists the number of training/testing samples.

The IP database was obtained by the airborne visible infrared imaging spectrometer (AVIRIS) instrument over Northwestern Indiana. The IP database consists of 145×145 hyperspectral pixels with 220 bands. There are 20 noisy bands on this database. The spectral wavelength ranges from 0.4 to 2.5 μm. This database is interested in 16 categories of land covers, most of which are crops. The number of training/testing samples is given in Table III.

The Salinas database was also gathered by the AVIRIS instrument. This database comprises 512×217 pixels, and each pixel has 224 bands. There are also 20 noisy bands on this database. The Salinas database includes 16 categories of land covers in farmlands. The number of training/testing samples is listed in Table IV.

[1] http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

TABLE II
Number of Training/Testing Samples in Each Category on the PU Database

Category	Name	Train	Test
1	Asphalt	100	6531
2	Meadows	100	18549
3	Gravel	100	1999
4	Trees	100	2964
5	Painted Metal Sheets	100	1245
6	Bare Soil	100	4929
7	Bitumen	100	1230
8	Self-Blocking Bricks	100	3582
9	Shadows	100	847
Total		900	41876

TABLE III
Number of Training/Testing Samples in Each Category on the IP Database

Category	Name	Train	Test
1	Alfalfa	5	41
2	Corn-Notill	143	1285
3	Corn-Mintill	83	747
4	Corn	24	213
5	Grass-Pasture	49	434
6	Grass-Trees	73	657
7	Grass-Pasture-Mowed	3	25
8	Hay-Windrowed	48	430
9	Oats	2	18
10	Soybean-Notill	98	874
11	Soybean-Mintill	246	2209
12	Soybean-Clean	60	533
13	Wheat	21	184
14	Woods	127	1138
15	Buildings-Grass-Trees-Drives	39	347
16	Stone-Steel-Towers	10	83
Total		1031	9218

TABLE IV
Number of Training/Testing Samples in Each Category on the Salinas Database

Category	Name	Train	Test
1	Brocoli Green Weeds 1	100	1909
2	Brocoli Green Weeds 2	100	3626
3	Fallow	100	1876
4	Fallow Rough Plow	100	1294
5	Fallow Smooth	100	2578
6	Stubble	100	3859
7	Celery	100	3479
8	Grapes Untrained	100	11171
9	Soil Vinyard Develop	100	6103
10	Corn Senesced Green Weeds	100	3178
11	Lettuce Romaine 4wk	100	968
12	Lettuce Romaine 5wk	100	1827
13	Lettuce Romaine 6wk	100	816
14	Lettuce Romaine 7wk	100	970
15	Vinyard Untrained	100	7168
16	Vinyard Vertical Trellis	100	1707
Total		1600	52529

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To quantitatively evaluate the classification performance of CASSN, this article employs overall classification accuracy (OA), average classification accuracy (AA), kappa, and accuracy under each category as evaluation measures [25]. Furthermore, to qualitatively measure the performance of CASSN, classification maps are also illustrated.

B. Experiment Settings

This section mainly introduces the data preprocessing of HSI and the hyperparameter settings of all methods. First, following the literature [49], the min–max scaling is leveraged to normalize the values of each band in HSI to the range [0, 1]. Then, HSI patches centered on the training/testing pixels are cropped from the normalized HSI. Each pixel in HSI has a corresponding HSI patch, and the spatial size of patches is 9×9 for all methods in experiments. Following the literature [29], to evaluate the generalization performance of CASSN and compared methods for HSI rotation, each cropped HSI patch is randomly rotated by 0°, 90°, 280°, or 270° for training and testing.

In experiments, the iterative epochs and the batch size of CASSN are set to 200 and 100, respectively. The learning rate of CASSN is initialized to 0.01 and then divided by 2 in every 20 iterative epochs. Compared methods utilize the same setting as in original literature.

C. Effectiveness of CEA Component

This section mainly studies the effectiveness of the CEA component. Recently, many SAMs [19], [23], [49], [50], [61] have been proposed to suppress redundant spectral bands for HSI classification. These SAMs usually generate the band weights based on the whole neighboring pixels in the HSI patch to recalibrate the importance of spectral bands. However, the whole pixels in the HSI patch may belong to different categories, not just the category of the center pixel [25]. Other categories of pixels in the HSI patch can interfere with the band weights since different categories of pixels usually have different redundant bands [49]. In this article, a CEA component is proposed to directly generate the band weights based on the pixel to be classified, which can effectively avoid interference from other categories of pixels in the HSI patch. In addition, both local information and global information of spectrums of the pixel are explored to generate the band weights. Then, the generated band weights are utilized to simultaneously recalibrate the importance of bands in the pixel and the HSI patch.

In this section, experiments are carried out on the PU database to verify the effectiveness of the CEA component. Since the CEA component directly affects the importance of spectral bands in the pixel and the HSI patch to fully study whether the CEA component can improve the discrimination of the pixel and the HSI patch, we conduct experiments without introducing any feature extraction component. First, a basic experiment is used as a reference to demonstrate the performance of experimental results. In the basic experiment, no SAM is employed. The original pixel and the HSI patch are directly fed to a softmax classifier for HSI classification, respectively. Second, the SAM proposed in the literature [49] is employed as a comparison. The SAM is utilized to generate the band weight based on the whole neighboring in the HSI patch to recalibrate the importance of spectral bands in the pixel and the HSI patch, respectively. Third, the local band weight in CEA is employed to recalibrate the importance of bands in the pixel and the HSI patch. Fourth, the global band weight in CEA is employed to recalibrate the importance of bands in the pixel and the HSI patch. Finally, the CEA is employed to recalibrate the importance of bands in the pixel and the HSI patch. These recalibrated pixels are directly fed to a softmax classifier for HSI classification. These recalibrated HSI patches are first fed to a global pooling layer for dimensionality reduction and then fed to a softmax classifier for HSI classification.

TABLE V
OA (%) OF DIFFERENT KINDS OF SPECTRAL ATTENTION COMPONENTS ON THE PU DATABASE

Components	Pixel	HSI Patch
Basic Experiment	75.84	85.99
SAM [49]	83.80	87.31
Local Band Weight	77.66	87.18
Global Band Weight	86.07	93.77
Cross Spectral Attention	**86.36**	**94.18**

Experiments are reported in Table V. Comparing the results of the "Basic Experiment" with other experiments, we can find that spectral attention can, indeed, improve the discrimination of the pixel and the HSI patch. SAM explores whole pixels in the HSI patch to generate the band weight. The HSI patch recalibrated by SAM achieves 87.31% accuracy. The pixel recalibrated by SAM achieves 83.80% accuracy, which is 2.56% lower than the accuracy of the pixel recalibrated by the proposed CEA. This demonstrates that the band weight generated by SAM cannot fully assess the importance of different bands in the pixel. The HSI patch recalibrated by the proposed CEA achieves 94.18% accuracy, which is 6.87% higher than the accuracy of the HSI patch recalibrated by SAM. This demonstrates that other categories of pixels in the HSI patch do interfere with the band weight, and the proposed CEA can effectively alleviate this problem. The performance of local band weight and global band weight demonstrates that both local information and global information of spectrums are important for generating effective band weights.

The band weights learned by the CEA are further visualized in Fig. 3 to show the effectiveness of CASSN. As shown in Fig. 3, the values of band weights in different bands are different, which demonstrates that learned band weights can effectively highlight informative bands for classification. In addition, the learned band weights of categories with similar spectrums (e.g. "Asphalt," "Gravel," "Bitumen," and "Self-Blocking Bricks") are also similar. The learned band weights can provide clues to distinguish these similar spectrums.

D. Ablation Study

The proposed CASSN contains three key components: the CEA component, the SFE component, and the CAA component. The performance of these components is studied on the PU database. Classification results are listed in Table VI.

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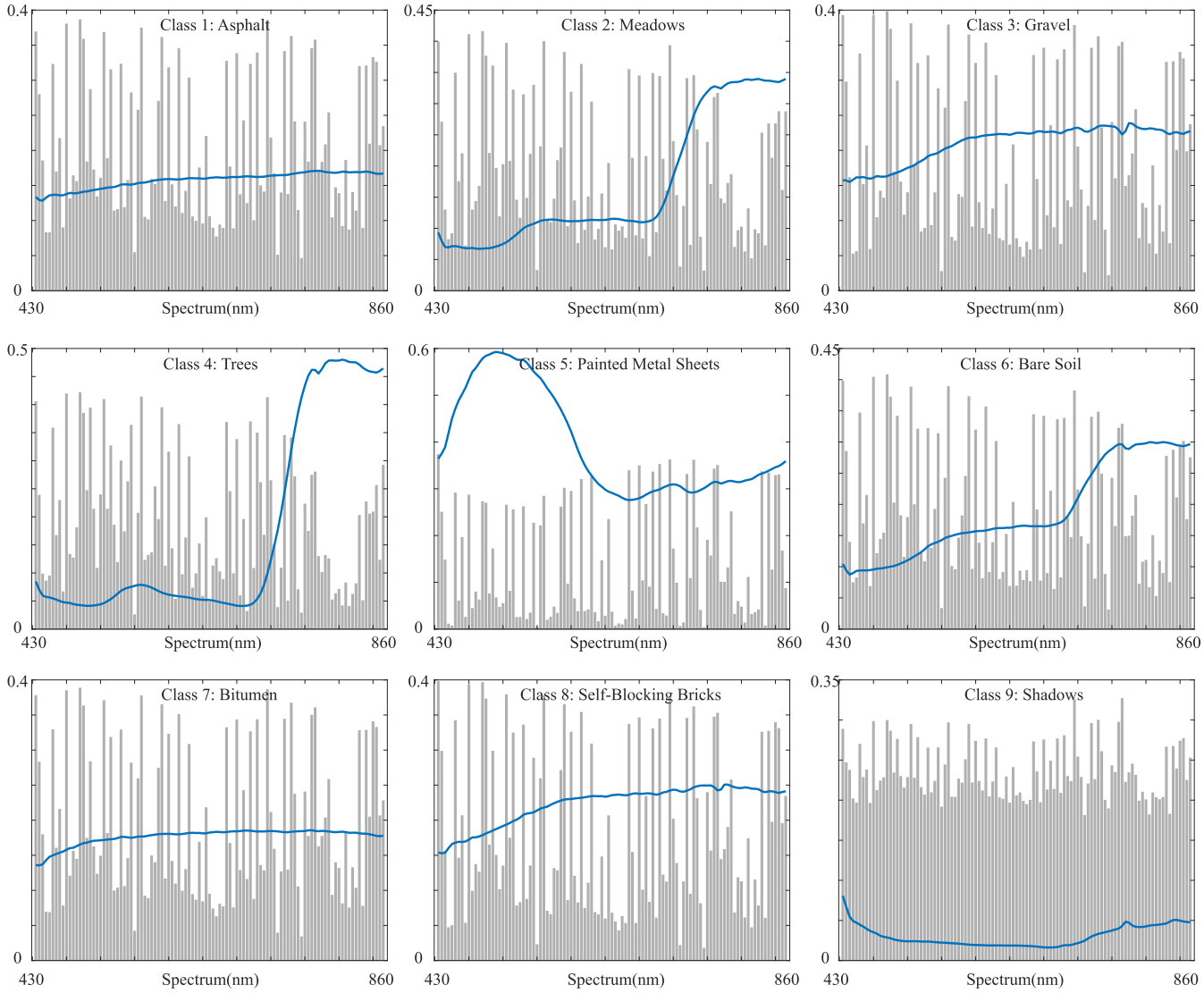


Fig. 3. Average spectrums (blue line) and average band weights (gray bar) in each category on the PU database.

CEA-SFE-Net represents the proposed CASSN without CAA. CEA-CAA-Net represents the proposed CASSN without SFE. SFE-CAA-Net represents the proposed CASSN without CEA. CEA-Net represents the proposed CASSN without SFE and CAA. SFE-Net represents the proposed CASSN without CEA and CAA. CAA-Net represents the proposed CASSN without CEA and SFE. Pixel-Patch represents that the pixel and the HSI patch are directly utilized for HSI classification. In CEA-SFE-Net, CEA-Net, SFE-Net, and Pixel-Patch, the global average pooling is utilized to reduce the dimensionality of features from the patch branch, and the concatenation operation is utilized to integrate the features from the pixel branch and the features from the patch branch. Then, a softmax classifier is utilized to obtain the category. As shown in Table VI, these three components are essential to improve the performance of CASSN.

TABLE VI
ABLATION STUDIES OF CASSN ON THE PU DATABASE

	CEA	SFE	CAA	OA (%)
CASSN	✓	✓	✓	**97.58**
CEA-SFE-Net	✓	✓		96.47
CEA-CAA-Net	✓		✓	96.04
SFE-CAA-Net		✓	✓	96.23
CEA-Net	✓			94.38
SFE-Net		✓		95.16
CAA-Net			✓	95.81
Pixel-Patch				87.98

E. Comparing With Other Methods

The performance of CASSN is further evaluated on PU, IP, and Salinas databases, respectively. Recent CNN- and GNN-based methods are utilized as comparison, including 1-D CNN [44], 2-D CNN [46], deformable HSI classification network (DHCNet) [62], squeeze multibias network (SMBN) [63], spectral attention network (SpecAN) [49], spectral–spatial attention network (SSAN) [25], minibatch graph convolutional network (miniGCN) [56], and graph fusion network (GFNet) [43]. In addition, a traditional feature extraction-based method, orthogonal total variation component analysis (OTVCA) [35], is also employed as a comparison.

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TABLE VII

ACCURACY IN EACH CATEGORY, OA, AA, AND KAPPA OF DIFFERENT METHODS ON THE PU DATABASE

Class	OTVCA	1-D CNN	2-D CNN	DHCNet	SMBN	SpecAN	SSAN	miniGCN	GFNet	CASSN
1	97.37	73.39	91.26	92.41	92.77	95.25	94.15	71.43	93.37	**99.10**
2	95.09	79.56	88.38	93.59	94.51	95.27	94.57	94.79	96.52	**96.56**
3	90.80	79.54	79.29	88.99	84.84	90.60	92.65	83.29	85.28	**98.85**
4	97.50	93.35	98.18	97.57	98.11	98.28	97.87	92.07	97.54	**99.49**
5	99.84	99.28	**100.00**	99.84	**100.00**	**100.00**	**100.00**	99.20	**100.00**	99.84
6	97.59	75.63	84.13	97.10	96.25	**98.15**	93.79	77.26	95.47	98.11
7	95.20	92.44	87.32	95.04	91.95	96.02	98.54	96.26	92.86	**99.92**
8	94.78	82.08	75.63	93.44	90.37	94.61	**96.45**	74.62	89.90	94.92
9	99.88	99.88	99.65	100.00	99.88	100.00	99.88	**100.00**	**100.00**	**100.00**
OA	95.92	80.70	88.04	94.23	94.08	95.80	95.10	86.90	94.92	**97.58**
AA	96.45	86.13	89.31	95.33	94.30	96.46	96.43	87.66	94.55	**98.53**
Kappa	94.60	75.06	84.29	92.38	92.16	94.44	93.51	82.56	93.28	**96.79**

TABLE VIII

ACCURACY IN EACH CATEGORY, OA, AA, AND KAPPA OF DIFFERENT METHODS ON THE IP DATABASE

Class	OTVCA	1-D CNN	2-D CNN	DHCNet	SMBN	SpecAN	SSAN	miniGCN	GFNet	CASSN
1	90.24	65.85	24.39	34.15	51.22	75.61	39.02	24.39	**100.00**	**100.00**
2	**95.10**	72.45	70.43	90.82	88.87	88.48	91.21	58.83	86.39	93.23
3	**92.37**	61.58	81.53	85.54	78.05	89.83	91.03	58.90	82.43	**92.37**
4	77.46	60.09	85.45	84.98	73.24	75.59	78.40	47.42	82.14	**93.90**
5	94.70	88.48	77.88	93.09	90.09	94.70	96.31	89.63	92.86	**98.16**
6	99.09	95.13	95.59	96.35	95.89	97.72	98.63	95.74	98.80	**99.24**
7	72.00	40.00	32.00	96.00	24.00	96.00	32.00	4.00	50.00	**100.00**
8	99.77	97.91	98.14	99.77	93.26	99.07	**100.00**	97.44	**100.00**	99.30
9	66.67	50.00	38.89	44.44	22.22	66.67	22.22	0.00	**100.00**	88.89
10	90.16	67.39	70.71	86.96	78.26	87.99	89.02	70.82	83.75	**93.59**
11	97.37	80.31	89.45	96.38	96.60	97.06	94.88	61.61	87.71	**98.14**
12	90.43	69.61	53.47	82.74	77.11	85.74	79.92	66.04	**94.34**	94.18
13	97.28	91.85	98.91	97.28	91.30	96.20	99.46	95.11	**100.00**	98.91
14	98.07	94.90	93.06	98.59	96.75	99.03	96.75	88.84	97.67	**99.74**
15	76.66	60.52	65.71	85.30	87.90	90.20	89.34	42.07	71.88	**97.41**
16	96.39	93.98	96.39	91.57	95.18	96.39	**98.80**	95.18	88.89	**98.80**
OA	94.35	78.82	81.76	92.23	89.13	93.10	92.42	70.38	89.76	**96.55**
AA	89.61	74.38	73.25	85.25	77.50	89.77	81.06	62.25	88.55	**96.62**
Kappa	93.54	75.73	79.13	91.13	87.55	92.11	91.34	66.60	88.30	**96.07**

TABLE IX

ACCURACY IN EACH CATEGORY, OA, AA, AND KAPPA OF DIFFERENT METHODS ON THE SALINAS DATABASE

Class	OTVCA	1-D CNN	2-D CNN	DHCNet	SMBN	SpecAN	SSAN	miniGCN	GFNet	CASSN
1	99.58	98.59	99.74	99.63	99.53	99.95	99.74	98.95	**100.00**	**100.00**
2	98.76	99.61	97.74	99.92	99.67	99.31	99.72	99.83	**100.00**	99.94
3	99.41	99.47	98.93	98.19	97.97	**100.00**	99.52	99.57	**100.00**	99.52
4	99.61	99.69	95.83	99.77	99.30	99.46	99.30	99.69	99.31	**99.85**
5	97.44	97.83	98.76	99.30	98.29	98.33	98.76	98.18	**100.00**	98.91
6	99.48	99.79	99.48	99.87	99.61	99.97	**100.00**	99.84	**100.00**	99.97
7	97.93	99.57	99.31	99.31	99.48	99.80	99.83	99.91	**100.00**	99.97
8	88.29	73.82	72.36	83.62	86.23	83.33	87.11	79.67	87.39	**91.04**
9	98.30	99.00	93.64	99.43	94.99	99.44	99.92	99.95	99.84	**99.97**
10	91.76	91.25	95.12	96.98	94.68	98.40	96.85	92.89	98.76	**99.40**
11	97.93	96.28	97.83	**100.00**	98.97	99.07	99.28	96.90	**100.00**	99.28
12	**100.00**	99.95	98.25	99.23	99.84	99.67	**100.00**	99.95	**100.00**	**100.00**
13	98.65	99.51	**100.00**	99.75	99.63	99.75	99.75	99.14	**100.00**	**100.00**
14	98.97	94.74	98.76	99.90	99.38	99.79	99.90	93.30	99.04	**99.90**
15	**95.88**	71.04	87.05	91.16	82.23	88.66	86.70	68.04	85.51	92.41
16	98.83	93.91	95.78	97.72	98.18	98.48	96.95	98.42	**100.00**	99.59
OA	95.70	89.21	90.61	94.77	93.37	94.50	94.99	90.45	95.40	**96.91**
AA	97.55	94.63	95.54	97.74	96.75	97.71	97.71	95.27	98.12	**98.73**
Kappa	95.21	87.98	89.57	94.18	92.61	93.88	94.42	89.35	94.89	**96.55**

Detailed results of CASSN on PU, IP, and Salinas databases are introduced as follows.

1) PU Database: First, the PU database is utilized to evaluate the performance of CASSN and compared methods. Quantitative experimental results are shown in Table VII. 1-D CNN that is a pixelwise method achieves different accuracies in different categories. In the "Painted Metal Sheets" and "Shadows" categories, 1-D CNN shows superior performance and achieves 99.28% and 99.88% accuracies. This shows that the spectrums of pixels in several categories are highly discriminative, and the spectrums of the pixel to be classified should be fully explored for HSI classification. When the spatial information of the HSI patch is introduced, the accuracies of compared methods in many categories have been increased, indicating that the spatial information is complementary to the spectral features and can effectively improve the classification

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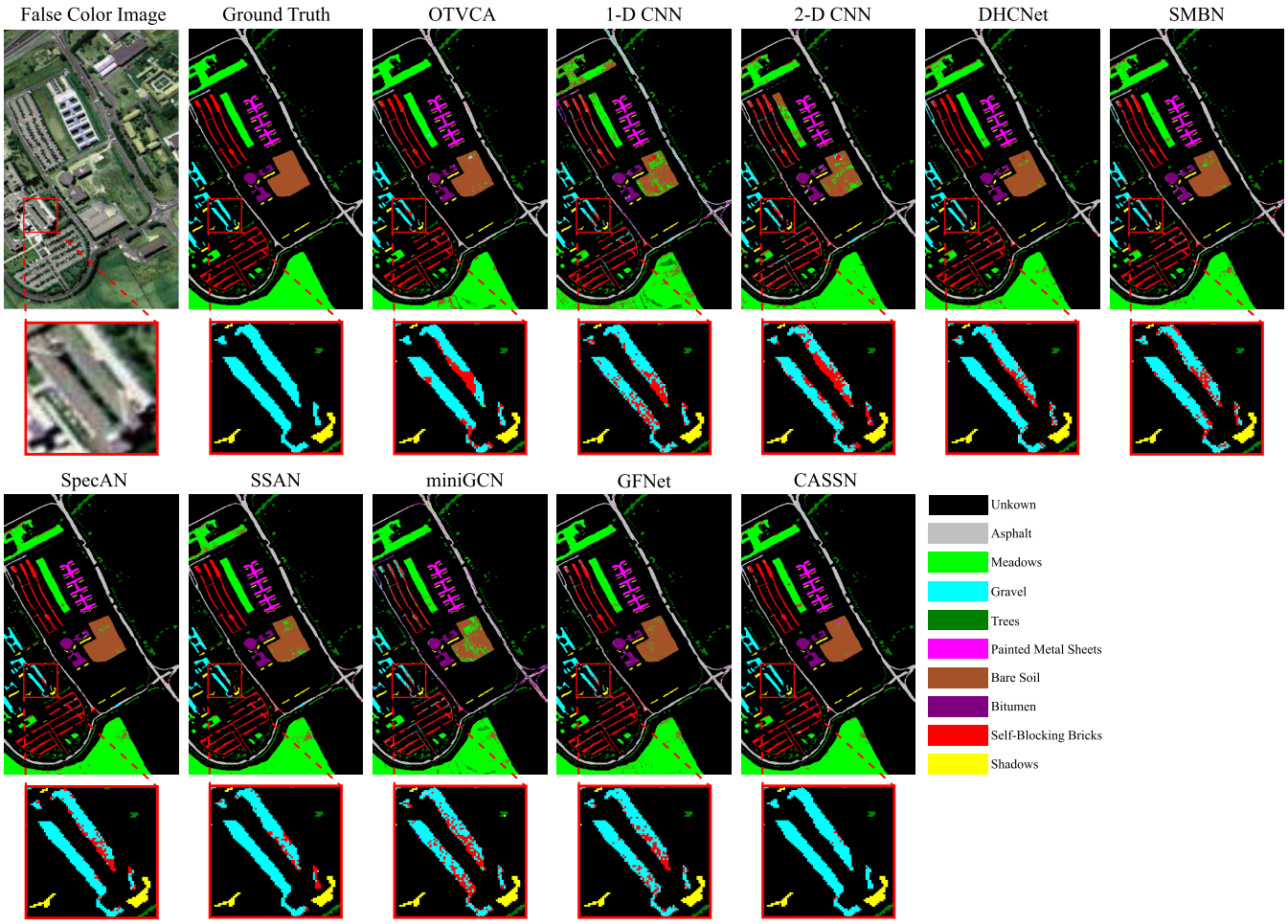


Fig. 4. False color image, ground truth, training pixels, and classification maps of CASSN and compared methods on the PU database.

results. However, the accuracies of 2-D CNN and SMBN are lower than that of 1-D CNN on the "Bitumen" category. This shows that it is necessary to design a more appropriate spatial feature extraction component. SpecAN, SSAN, and CASSN that are attention-based methods can extract effective spatial features, which show better performance than 1-D CNN on the "Bitumen" category. The proposed CEA component can effectively suppress redundant bands, and the proposed CAA can extract the discriminative spectral–spatial feature that is robust to the HSI rotation. Due to the effectiveness of the CEA component and the CAA component, the proposed CASSN outperforms compared methods.

The false color image, ground truth, training pixels, and visual classification maps on the PU database are shown in Fig. 4. Since 1-D CNN cannot effectively explore spatial information, the classification map of 1-D CNN has a lot of noise. From the classification maps of DHCNet, SMBN, SpecAN, and SSAN, it can be seen that the pixels misclassified are mainly distributed near the edges of various land-cover regions. There are two reasons for the poor performance of these methods: on the one hand, because there are different categories of pixels in the HSI patch, it will interfere with the generation of band weights. On the other hand, near the region edges, the HSI patches may have contours in various directions, which is the same problem caused by HSI rotation. The 2-D and 3-D convolutions fail to extract robust spatial features from these HSI patches. The classification map of CASSN is relatively smooth, and there are better classification results near the edges of various land-cover regions, which can demonstrate the effectiveness of CASSN.

2) IP Database: The performance of CASSN is further evaluated on the IP database. The quantitative performance on the IP database is reported in Table VIII. The number of training pixels in the "Alfalfa," "Grass-Pasture-Mowed," and "Oats" categories is very small, which makes the HSI classification challenging. 1-D CNN achieves 65.85%, 40.00%, and 50.00% in the "Alfalfa," "Grass-Pasture-Mowed," and "Oats" categories, respectively. The performance of 2-D CNN, SMBN, and SSAN is far weaker than that of 1-D CNN in the "Alfalfa," "Grass-Pasture-Mowed," and "Oats" categories. This shows that too few training samples will make it difficult for 2-D CNN, SMBN, and SSAN to learn effective spatial features. The proposed CASSN outperforms compared methods in most categories on the IP database.

The false color image, ground truth, training pixels, and visual classification maps on the IP database are displayed in Fig. 5. Compared methods still show poor performance near the region edges on the IP database. The classification map of 1-D CNN is not smooth. On the IP database, the

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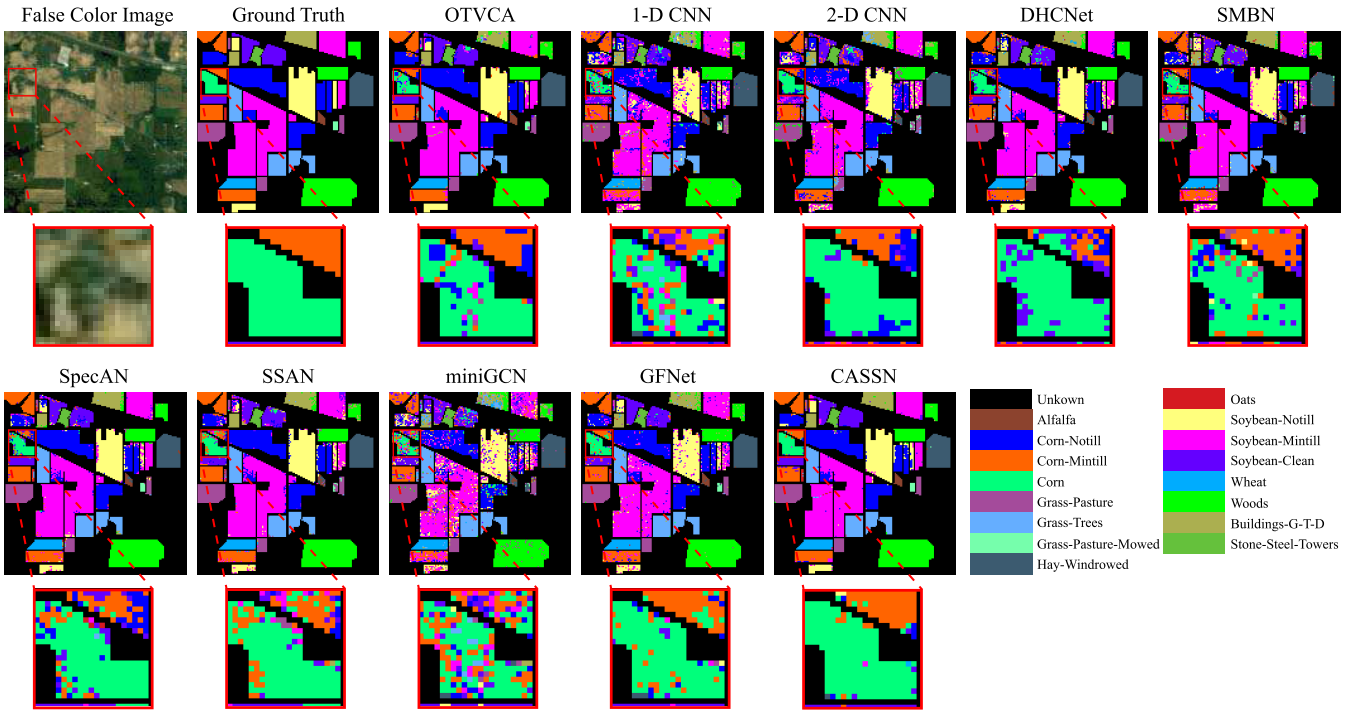


Fig. 5. False color image, ground truth, training pixels, and classification maps of CASSN and compared methods on the IP database.

region shapes of several land covers ("Alfalfa," "Grass-Pasture-Mowed," and "Oats") are narrow. Compared methods show poor performance in these narrow land covers. Although the classification maps of CASSN are smooth, the pixels misclassified by CASSN are also mainly concentrated at the region edges. This shows that it is very challenging to classify pixels near the region edges. The effective use of the spatial information contained in HSI needs further research.

3) Salinas Database: The Salinas database is further employed to demonstrate the performance of CASSN. Table IX reports quantitative experimental results on the Salinas database. The Salinas database mainly includes different crops. Since the spectrums of several kinds of crops are quite different, several crops can be distinguished only by using the spectrums. The accuracies of 1-D CNN on eight categories of land covers, "Brocoli Green Weeds 2," "Fallow," "Fallow Rough Plow," "Stubble," "Celery," "Soil Vinyard Develop," "Lettuce Romaine 5wk," and "Lettuce Romaine 6wk," even exceed 99%. In the Salinas database, since the spatial information is introduced, DHCNet, SMBN, SpecAN, and SSAN show better performance than 1-D CNN in each category of land covers. The proposed CASSN outperforms compared methods in 12 categories of land covers.

The false color image, ground truth, training pixels, and visual classification maps on the Salinas database are displayed in Fig. 6. At the edges of the "Grapes Untrained" category, compared methods show poor performance. The classification map of 1-D CNN in the "Grapes Untrained" and "Vinyard Untrained" categories contains a lot of noise. Since the spectrums of the "Grapes Untrained" and "Vinyard Untrained" categories are very similar, the proposed CASSN and compared methods are easy to confuse these two categories of pixels. The results in Fig. 6 show that the spectral–spatial features of CASSN cannot be utilized to effectively distinguish these two categories. It is still necessary to further study how to design a discriminative spatial feature extractor to distinguish these two categories. As shown in Fig. 6, the proposed CASSN has smoother classification maps than compared methods.

TABLE X
OAs (%) OF CASSN WITH DIFFERENT SIZES OF HSI PATCHES

	5×5	7×7	9×9	11×11	13×13
PU	96.16	96.82	97.58	97.61	97.70
IP	95.83	96.28	96.55	96.52	96.62
Salinas	94.19	95.90	96.91	97.41	97.89

F. Effect of the Size of HSI Patches

Large-size HSI patches will introduce more spatial information, which can affect the performance of CASSN. The performance of CASSN with different sizes of HSI patches is explored in this section. OAs of CASSN with 5×5 to 13×13 HSI patches are reported in Table X. As shown in Table X, the proposed CASSN will have better results when using large-size HSI patches for classification. In PU and IP databases, when the size of HSI patches exceeds 9×9, the performance improvement rate brought by the larger size gradually decreases.

G. Effect of the Number of Training Samples

In remote sensing, it is very expensive to label numerous training samples. We further study the performance of CASSN under different numbers of training samples. Since the number of samples in different categories on the IP database is unbalanced, and the number of samples in several categories is only more than 20, the IP database is not convenient for experiments

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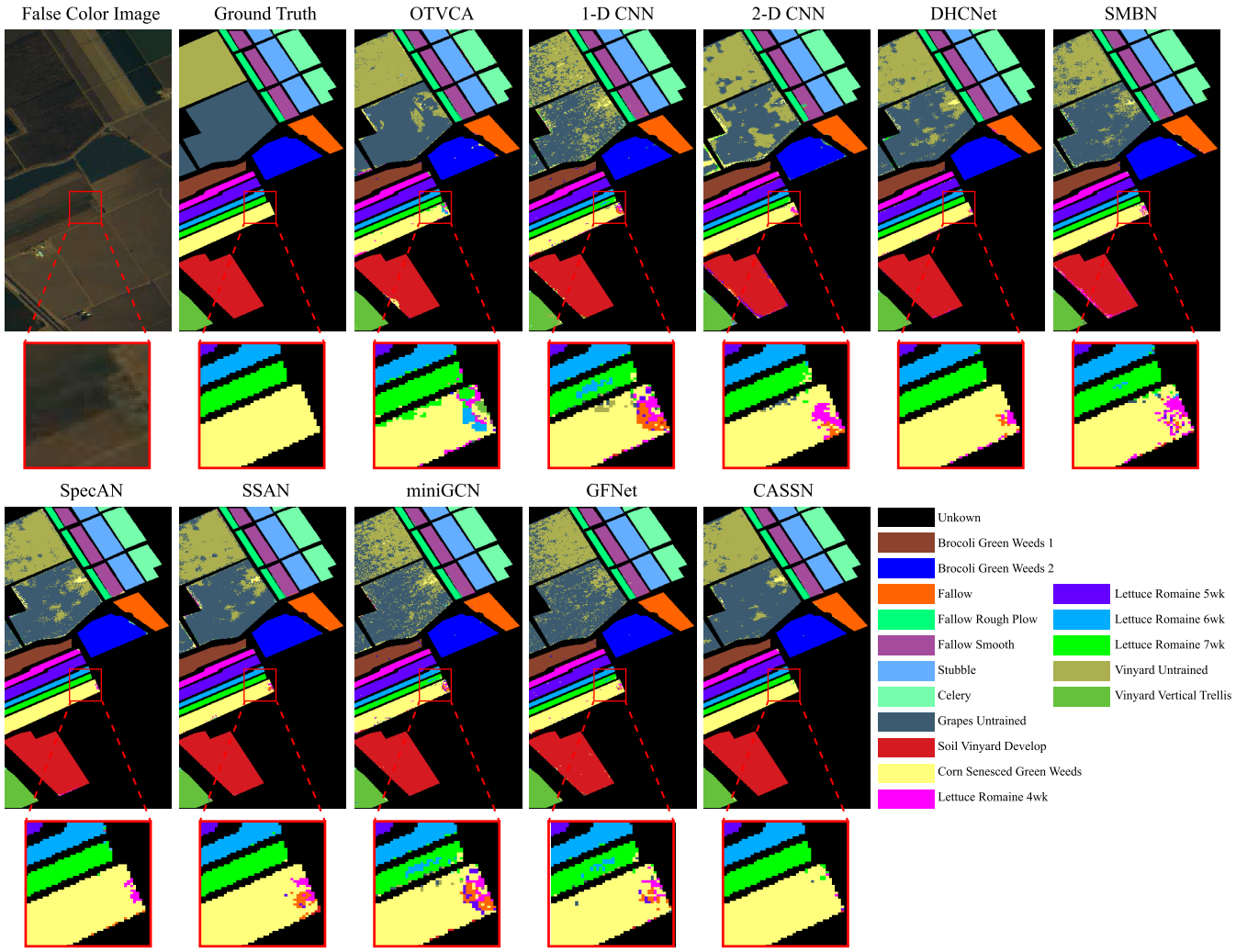


Fig. 6. False color image, ground truth, training pixels, and classification maps of CASSN and compared methods on the Salinas database.

TABLE XI
OAs (%) OF CASSN WITH DIFFERENT NUMBERS OF TRAINING SAMPLES IN EACH CATEGORY

	10	25	50	75	100	150	200
PU	72.75	86.92	92.58	95.67	97.58	98.59	98.79
Salinas	85.60	91.66	95.64	95.93	96.91	97.40	98.22

with different sample sizes. The experiments are conducted on PU and Salinas databases. The number of training samples in each category is set to 10, 25, 50, 75, 100, 150, and 200 for experiments, respectively. The OAs of CASSN with different numbers of training samples are listed in Table XI. Since the proposed CASSN is not specially designed for the small sample problem, when the number of training samples is 10, the performance of the proposed CASSN decreases greatly. As the number of samples increases, the performance of CASSN increases significantly. In the future, we will study new lightweight convolutional networks to deal with the small sample problem.

H. Computational Cost

The number of weight parameters and floating-point operations (FLOPs) of CASSN is related to the number of bands and land-cover categories on the databases. Since the number of bands and land-cover categories of IP and Salinas databases is more than that of the PU database, the number of weight parameters and FLOPs of CASSN on IP and Salinas databases is more than on the PU database. On the PU, IP, and Salinas databases, the numbers of weight parameters of CASSN are 49 467, 102 682, and 104 970, respectively. On the PU, IP, and Salinas databases, the numbers of FLOPs of CASSN are approximately 3.20M, 4.48M, and 4.53M, respectively.

TABLE XII
TRAINING AND TESTING TIME (SECONDS) OF EACH METHOD

	PU		IP		Salinas	
	Train	Test	Train	Test	Train	Test
OTVCA	182.71	1.47	27.95	0.23	131.80	0.81
1D-CNN	35.23	3.51	70.81	1.47	109.61	8.30
2D-CNN	40.85	4.04	75.74	1.67	122.56	9.51
DHCNet	80.21	9.34	121.80	2.62	192.32	14.44
SMBN	67.82	5.37	107.54	1.86	167.21	11.15
SpecAN	47.48	4.41	83.17	1.68	133.30	9.89
SSAN	326.74	23.08	738.43	10.63	1160.25	59.06
miniGCN	241.84	11.92	82.49	8.36	476.74	19.07
GFNet	273.03	12.57	214.37	9.06	522.14	22.95
CAASN	53.22	4.35	88.33	1.72	141.25	9.82

The time cost of different methods is reported in Table XII. In this article, all deep learning-based methods are

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reimplemented with TensorFlow and run on a workstation equipped with Intel Core i7-5930K, 64-GB RAM, and NVIDIA Titan X. The training and testing times of CASSN are close to that of 2-D-CNN and SpecAN. The proposed CEA and the CAA modules are based on simple convolution operations and do not significantly add the time cost. The proposed CASSN not only has good classification performance but also has less time-consuming.

V. Conclusion

This article proposes a CASSN to alleviate the problem that previous methods are not robust to HSI rotation. First, a CEA is proposed to fully explore the local information and global information of the spectrums in the pixel to simultaneously recalibrate the importance of bands in the pixel and the HSI patch. Second, an SFE is designed to extract features from the pixel and the HSI patch in the spectral dimension. Finally, a CAA is proposed to integrate the features of the pixel and the HSI patch to generate the spectral–spatial feature, which is robust to the HSI rotation. The proposed CASSN outperforms compared methods on three public databases. However, the proposed CASSN still shows poor performance near the region edges on the IP database. In the future, we will study new convolutional networks to elaborately leverage the spatial information of HSI for classification.

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